

Entrepreneurship, Frictions, and Wealth

May 10, 2026

Rudan Zheng

Abstract

Cagetti and De Nardi (2006) study how entrepreneurship, borrowing constraints, and bequest motives help explain the high concentration of wealth in the United States. Their model combines a life-cycle household problem with occupational choice: households can work for wages or operate a business, and entrepreneurs choose both saving and business capital while facing an enforcement constraint on borrowing. The original paper calibrates this environment to match key wealth and entrepreneurship moments and finds that entrepreneurs are central for explaining top wealth shares, while tighter borrowing constraints reduce both inequality and aggregate entrepreneurial activity. In this reproduction paper, I focus on the theoretical model rather than the SCF data work or the full general equilibrium calibration. I restate the household problem, decompose the worker and entrepreneur branches, run the existing theory notebook, and compare the notebook's outputs with the benchmark results from the paper.

Keywords entrepreneurship, wealth inequality, borrowing constraints, occupational choice, bequests, reproduction

JEL codes D31, E21, J23

Powered by Econ-ARK

GitHub Repo: <https://github.com/econ-ark/Cagetti2005tk>
html: <https://econ-ark.github.io/Cagetti2005tk/>
PDF: [Cagetti2005tk.pdf](#)

Original article: Cagetti and De Nardi (2006), Journal of Political Economy. DOI: 10.1086/508032

This project follows the course reproduction format and focuses on the theoretical household problem from Cagetti and De Nardi (2006). The authors would like to thank seminar participants at the Board of Governors of the Federal Reserve System, Deutsche Bundesbank, the European Central Bank, the Norwegian University of Science and Technology, Oslo Macro Group, the Society for Computational Economics in 2023 and 2024, Statistics Norway, the University of Pennsylvania, Vanderbilt University, Johns Hopkins University, and West Virginia University for valuable feedback. The authors would also like to thank Sasha Indarte, Adrian Monninger, Plutarchos Sakellaris, three anonymous referees, and the editor Morten Ravn for insightful comments on earlier versions of the paper. Thanks also to the <https://github.com/econ-ark/econ-ark.org> team members who helped with the project, particularly Alan Lujan, Akshay Shanker, and Matthew White (alphabetically).

1 Introduction

Cagetti and De Nardi (2006) study why wealth is so concentrated in the United States and why entrepreneurs are so important at the top of the wealth distribution. Their starting point is that entrepreneurs are a small share of households but own a large share of total wealth. A standard incomplete-markets saving model can explain precautionary saving, but it has difficulty producing the very thick upper tail of wealth. The original paper adds entrepreneurship, borrowing frictions, and voluntary bequests to address this problem.

The model is built around a household that chooses between two occupations. As a worker, the household earns labor income and saves at the risk-free rate. As an entrepreneur, the household operates a business technology, chooses business capital, and can potentially earn a higher return. The key friction is limited enforcement. An entrepreneur can borrow only if repaying the loan is better than defaulting, keeping a fraction of working capital, and returning to work as a wage earner. Because of this constraint, personal wealth matters as collateral. Richer entrepreneurs can run larger firms, and successful entrepreneurs have a strong incentive to keep saving.

The original paper calibrates this environment to match moments such as the capital-output ratio, the fraction of entrepreneurs, transition rates into and out of entrepreneurship, relative wealth of entrepreneurs, and the wealth Gini coefficient. Its main quantitative message is that entrepreneurs and borrowing constraints help explain wealth concentration. More restrictive borrowing constraints reduce top wealth shares but also reduce firm size, aggregate capital, and the number of entrepreneurs. The paper also emphasizes voluntary bequests: dynastic saving helps some high-ability descendants enter or continue entrepreneurship.

My reproduction paper has a narrower goal. I reproduce and analyze the theoretical household problem using the existing notebook in this repository. I do not reproduce the SCF data cleaning, the full calibration, the complete stationary general equilibrium, or all counterfactual tables. Instead, I focus on whether the notebook correctly organizes the worker and entrepreneur branches, implements the enforcement constraint, solves the fixed-price household problem, and produces diagnostic results that can be compared with the original paper's benchmark results.

The rest of the paper is organized as follows. Section 2 summarizes the motivation from the original paper and the related entrepreneurship literature. Section 3 explains the model. Section 4 describes the computational decomposition used in the notebook. Section 5 organizes the parameters. Section 6 compares the reproduction outputs with the paper's benchmark results and discusses the figures. Section 7 concludes.

2 Wealth, Entrepreneurship, and Borrowing Constraints

The original paper is motivated by a simple empirical fact: entrepreneurs are a small part of the population but hold a large share of wealth. I use those facts as background for

the model reproduction, not as newly cleaned or newly reproduced SCF evidence in this project. The point for the current write-up is that the theory needs a mechanism that can connect business ownership, saving, and the upper tail of the wealth distribution.

Cagetti and De Nardi classify entrepreneurs using information on self-employment, business ownership, and active management. Table 1 reports their benchmark definitions. The narrowest definition, self-employed business owners, is closest to the model idea of an active entrepreneur because it excludes households that only passively own a business.

Table 1 Original Paper’s Entrepreneurship Definitions and Wealth Shares

Definition	Percent in population	Share of total wealth
Business owners or self-employed	16.7	52.9
All business owners	13.3	48.8
Active business owners	11.5	41.6
All self-employed	11.1	39.0
Self-employed business owners	7.6	33.0

The model also starts from the observation that wealth is much more concentrated than labor earnings. Table 2 gives the original paper’s summary of the top of the U.S. wealth distribution. In this reproduction I treat these numbers as benchmark facts from the paper, not as outcomes of my own empirical data pipeline.

Table 2 Benchmark Wealth Concentration Reported in the Original Paper

	Fraction of people, top			
	1%	5%	10%	20%
Total net worth held	30%	54%	67%	81%

The literature before Cagetti and De Nardi had already made two points that are important for the reproduction. First, incomplete-markets models can generate precautionary saving but often have trouble matching the extreme upper tail of wealth. Second, entrepreneurship models show that liquidity and borrowing constraints can affect entry, investment, and saving. Examples include Evans and Jovanovic (1989), Quadrini (2000), Gentry and Hubbard (2004), and Buera (2006). The contribution of the Cagetti and De Nardi model is to put occupational choice, entrepreneurial investment, borrowing constraints, and intergenerational wealth transmission inside one household problem.

For this project, the borrowing-constraint literature matters mainly because it tells me what to reproduce theoretically. The enforcement constraint is not just an exogenous debt limit. It is an incentive condition: the entrepreneur must prefer to repay rather than default and work as a wage earner. That is why the notebook gives special attention to the entrepreneur branch and to the comparison between the value of honoring debt and the value of default.

3 The Model

This section explains the theoretical model objects that I reproduce in the notebook. The aim is not to rewrite the original Cagetti and De Nardi paper, but to identify the pieces of the household problem that have to be represented in code: the state variables, the worker branch, the entrepreneur branch, the enforcement constraint, and the continuation values.

3.1 Household State and Timing

There is a continuum of households and no aggregate uncertainty. The full individual state can be written as

$$\mathbf{x} = (a, y, \theta, s),$$

where a is beginning-of-period assets, y is worker productivity, θ is entrepreneurial ability, and s records the household's demographic and occupational status. In the paper this status separates young workers, young entrepreneurs, old entrepreneurs, and old retirees. In the notebook I keep the same economic distinction through the value functions V , V_w , V_e , W , W_e , and W_r .

The timing is useful to state explicitly. At the start of the period, the household observes assets and current abilities. It then makes a discrete status decision: a young household chooses between working and entrepreneurship, while an old entrepreneur chooses between continuing the business and retiring. After that branch is chosen, the household chooses continuous controls. Everyone chooses consumption c and next-period assets a' . Entrepreneurs also choose working capital k , which is an intra-period business scale and does not itself carry into the next period as a state variable.

3.2 Preferences, Demographics, and Continuation

Households have CRRA period utility,

$$u(c) = \frac{c^{1-\sigma}}{1-\sigma}. \tag{1}$$

Future utility is discounted by β . The model also includes intergenerational altruism through the parameter η . When $\eta = 0$, the household has no voluntary bequest motive. When $\eta = 1$, the household values the descendant like a dynastic continuation.

Demographics are represented with two survival probabilities. A young household remains young with probability π_y and becomes old with probability $1 - \pi_y$. An old household survives with probability π_o and dies with probability $1 - \pi_o$, in which case a descendant enters the model with inherited assets. The notebook keeps these continuation weights in the Bellman equations, although its stationary distribution diagnostic is young-only rather than a full demographic distribution.

3.3 Worker Income and Entrepreneurial Technology

Workers earn labor income from their worker productivity:

$$a' = (1 + r)a + (1 - \tau)wy - c. \quad (2)$$

Here r is the risk-free interest rate, w is the wage, and τ is the payroll tax used to finance pensions. A retired household receives pension income p and solves a similar consumption-saving problem:

$$a' = (1 + r)a + p - c. \quad (3)$$

Entrepreneurs operate a decreasing-returns technology. If an entrepreneur invests working capital k , output is

$$\theta k^\nu,$$

with $\nu < 1$. The entrepreneur's resources after production and debt repayment are

$$a' = (1 - \delta)k + \theta k^\nu - (1 + r)(k - a) - c. \quad (4)$$

This equation is the main reason entrepreneurship is different from ordinary saving. The entrepreneur can choose a business scale k that may exceed assets a , but doing so requires borrowing and therefore must satisfy the enforcement constraint.

3.4 Borrowing and Enforcement

The model does not impose a simple fixed borrowing limit. Instead, borrowing is limited by imperfect contract enforcement. If an entrepreneur defaults, the entrepreneur keeps a fraction f of working capital and becomes a worker for one period. Lenders can seize the remaining fraction. Therefore a feasible entrepreneurial choice must make repayment at least as valuable as default.

For a young entrepreneur the enforcement condition is

$$u(c) + \beta\pi_y EV(a', y', \theta') + \beta(1 - \pi_y)EW(a', \theta') \geq V_w(fk, y, \theta). \quad (5)$$

The left side is the value of honoring the contract. The right side is the value of defaulting, keeping fk , and entering the worker branch. A larger f means default is more attractive, so the borrowing constraint becomes tighter. In the notebook this condition is implemented by checking each candidate k against the relevant default value.

3.5 Worker and Entrepreneur Branches

The young household first compares the worker and entrepreneur branches:

$$V(a, y, \theta) = \max\{V_w(a, y, \theta), V_e(a, y, \theta)\}. \quad (6)$$

The worker branch is a standard consumption-saving problem:

$$V_w(a, y, \theta) = \max_{c, a'} \{u(c) + \beta \pi_y EV(a', y', \theta') + \beta(1 - \pi_y) W_r(a')\}, \quad (7)$$

subject to equation (2) and $a' \geq 0$. If a young worker ages, the model sends the household to retirement, so the continuation term uses W_r .

The entrepreneur branch adds the business capital choice:

$$V_e(a, y, \theta) = \max_{c, k, a'} \{u(c) + \beta \pi_y EV(a', y', \theta') + \beta(1 - \pi_y) EW(a', \theta')\}, \quad (8)$$

subject to equation (4), $a' \geq 0$, $k \geq 0$, and the enforcement condition (5). This is the branch where the reproduction has to do the most work, because k affects both business returns and the default value.

3.6 Old Households, Retirement, and Bequests

Old entrepreneurs choose whether to continue operating the business or retire:

$$W(a, \theta) = \max\{W_e(a, \theta), W_r(a)\}. \quad (9)$$

The continuing entrepreneur branch is similar to the young entrepreneur branch, but the continuation value reflects survival as old and possible death with a descendant:

$$W_e(a, \theta) = \max_{c, k, a'} \{u(c) + \beta \pi_o EW(a', \theta') + \eta \beta(1 - \pi_o) EV(a', y', \theta')\}. \quad (10)$$

The old entrepreneur faces the same resource equation and a default comparison against retirement value $W_r(fk)$.

Retirement is absorbing in the sense that a retired old household does not re-enter entrepreneurship:

$$W_r(a) = \max_{c, a'} \{u(c) + \beta \pi_o W_r(a') + \eta \beta(1 - \pi_o) EV(a', y', \theta')\}, \quad (11)$$

subject to equation (3). This equation is also where voluntary bequests enter. When old households die, descendants receive assets and draw ability states, so saving can matter for both the current household and the next generation.

3.7 Stationary Equilibrium

The full paper closes the model in stationary general equilibrium. A stationary equilibrium consists of prices (r, w) , a tax rate τ , policy functions for consumption, saving, occupation, and entrepreneurial investment, and an invariant distribution over household states. Prices are consistent with the nonentrepreneurial production sector, the social-security budget balances, and the distribution is unchanged by the household transition rules.

For this reproduction, I only use that equilibrium definition to understand what would be needed for a full replication. The current notebook fixes prices and policy inputs, solves the household problem, and reports diagnostics from that partial-equilibrium solution. Therefore the theory reproduction reconstructs the main dynamic program, but it does not claim to reproduce the paper's full stationary general equilibrium.

4 Computational Decomposition

The purpose of the decomposition is not to introduce a new economic model. It is only a way to make the Cagetti and De Nardi household problem easier to solve and inspect in a notebook. The original model combines saving, occupational choice, entrepreneurial investment, borrowing constraints, survival, and bequests in one dynamic problem. I separate these objects into a small number of computational blocks so that each economic decision is visible.

The beginning-of-period state is assets plus the relevant ability variables. For a young household the state is (a, y, θ) . For an old entrepreneur it is (a, θ) , and for an old retiree it is just a . The first important computational step is a branch comparison. Young households compare the worker value V_w with the entrepreneur value V_e . Old entrepreneurs compare the continuing-business value W_e with retirement value W_r . These comparisons are pointwise maxima on the state grid.

The worker branch is closest to the standard consumption-saving problem. Given assets and labor income, the notebook builds worker cash-on-hand,

$$m^w = (1 + r)a + (1 - \tau)wy,$$

and then solves the choice between consumption and next-period assets. The continuation value already contains the expected next-period worker and entrepreneur values, so this branch can focus on the one continuous saving decision.

The entrepreneur branch is the key part of the reproduction because it adds business capital and the enforcement constraint. For each state, the notebook searches over candidate values of k . Each k implies entrepreneurial resources

$$m^e = (1 - \delta)k + \theta k^\nu - (1 + r)(k - a).$$

The notebook then solves the consumption-saving choice given m^e and keeps only those k choices for which honoring the debt is at least as valuable as default. In this way, the borrowing limit is not imposed as a fixed number; it is checked through the value functions themselves.

The continuation step is where the dynamic model comes back together. The notebook builds continuation-value functions as weighted sums of interpolated future values. For young households, the weights include the probability of staying young and the probability of aging. For old households, the weights include old-age survival and the descendant value that matters when the old household dies. The notebook also respects the timing detail that a young worker who ages enters retirement directly.

Table 3 Notebook Decomposition of the Household Problem

Block	Economic role	How it appears in the notebook
Beginning state	Carries assets and abilities into the period	Arrays over a , y , and θ for young households; arrays over a and θ for old entrepreneurs
Branch choice	Chooses the best occupational or retirement branch	Pointwise comparisons $V = \max\{V_w, V_e\}$ and $W = \max\{W_e, W_r\}$
Worker branch	Standard labor-income consumption-saving problem	Builds m^w and solves for consumption and a'
Entrepreneur branch	Chooses business scale and saving under enforcement	Discrete search over k with a default-value check
Continuation	Carries assets and ability transitions forward	Expected values include survival, aging, ability transitions, retirement, and descendant values
Diagnostics	Shows what the fixed-price solution implies	Figures for values, policies, k^* , convergence, and the young-only distribution check

Cell by cell, the notebook first states the Bellman system, then defines structural parameters, fixed-price inputs, grids, transition matrices, and helper interpolation functions. After that it defines the worker, entrepreneur, retiree, and continuation updates, runs value-function iteration, and produces diagnostic figures and summary tables. This organization is useful because a reader can inspect the economic role of each block instead of treating the whole household problem as one opaque maximization.

5 Parameter Organization

The parameter section in this reproduction has a limited purpose. I do not re-estimate the model and I do not clean the SCF data used in the original calibration. Instead, I organize the parameters that the notebook takes as inputs. This makes it clear which objects come from the paper’s calibration and which objects are numerical choices needed to solve the fixed-price household problem.

The first group is the economically meaningful parameter block. These are the parameters that define preferences, technology, demographics, ability processes, and the enforcement friction. The notebook’s `PaperStructuralParams` class keeps these objects in one place so that a reader can connect the code back to the Bellman equations.

The second group is specific to the current reproduction. The notebook works at fixed prices, using an interest rate close to the paper’s baseline solved economy and normalizing wages. The payroll tax and pension rule are included because they appear

Table 4 Parameters Used in the Theory Reproduction

Symbol	Description	Value used	Role in reproduction
<i>Structural parameters from the paper</i>			
σ	Risk aversion	1.5	Preferences
β	Discount factor	0.865	Intertemporal utility
δ	Depreciation rate	0.06	Business capital
ν	Decreasing returns	0.88	Entrepreneurial technology
f	Default retention share	0.75	Enforcement constraint
π_y	Prob. of staying young	0.978	Demographics
π_o	Prob. of old-age survival	0.911	Demographics
η	Altruism weight	1.0	Bequest motive
θ	Entrepreneurial ability grid	(0, 0.514)	Ability states
$\mathbf{P}_\theta$	Ability transition	See notebook	Ability persistence
$\mathbf{P}_y$	Labor-income transition	See notebook	Worker productivity
<i>Fixed-price inputs in the notebook</i>			
r	Net interest rate	0.065	Fixed price input
w	Wage normalization	1.0	Fixed price input
τ	Payroll tax	0.10	Illustrative input
p	Pension payment	$0.4 w E[y]$	Pension rule

in the worker and retiree budget constraints, but the notebook does not solve for the social-security budget balance. This is why I treat τ as an input rather than as a reproduced equilibrium outcome.

Finally, the notebook also has numerical settings that are not structural parameters: the asset grid, the grid for entrepreneurial capital k , value-function-iteration tolerances, damping, and the EGM stabilization option. These settings affect the approximation, but they are not part of the economic calibration. I keep them separate in the code because changing a grid should not be confused with changing the model.

6 Reproduction Results and Discussion

This section compares the notebook results with the benchmark results in Cagetti and De Nardi (2006). The comparison has to be careful because my notebook solves a fixed-price household problem and uses a young-only stationary distribution diagnostic. The original paper solves a full stationary general equilibrium with young and old households, bequests, market clearing, and calibration to data moments. Therefore, I treat the notebook results as a partial reproduction of the theory mechanism, not as a full replication of the quantitative tables.

6.1 Convergence and Computation

The first result is that the value-function iteration now converges. Figure 1 shows the total update norm and the separate update norms for the main value functions. The notebook reaches the stopping rule after 115 iterations, with a final sum norm of 4.78×10^{-4} against a tolerance of 5.0×10^{-4} . This is important because the later occupational-choice and distribution results are based on a converged fixed-price solution rather than on an unfinished iterate.

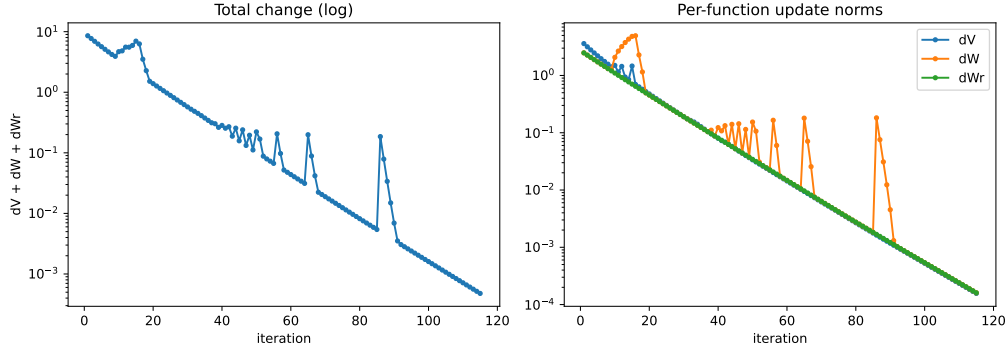


Figure 1 Convergence diagnostics for the fixed-price household problem.

The enforcement constraint is active in the computation. In the baseline run, all entrepreneurial candidate points are feasible in the resource sense, but only about 12.9 percent of young entrepreneur candidates pass the repayment-versus-default check. For old entrepreneurs the enforcement pass rate is about 67.0 percent. This difference is economically sensible: old high-ability entrepreneurs in the solved grid have a stronger incentive to continue operating firms, while many young households still find the worker branch more attractive.

6.2 Value Functions and Policy Functions

Figure 2 summarizes the most useful model objects from the notebook. The top-left panel compares the young worker value V_w and young entrepreneur value V_e for high entrepreneurial ability and middle labor productivity. The entrepreneur value is very low at the borrowing-constrained end of the grid, which reflects the difficulty of running a firm with too little collateral. At higher assets, the occupational envelope becomes much closer, and the branch choice is determined by the comparison between the worker outside option and entrepreneurial returns.

The top-right panel shows that old high-ability entrepreneurs prefer continuing entrepreneurship over retirement throughout the plotted asset grid. This matches the intuition of the original paper: once a high-ability entrepreneur has access to the business technology, continuing can be valuable because entrepreneurial returns are high relative to the safe return.

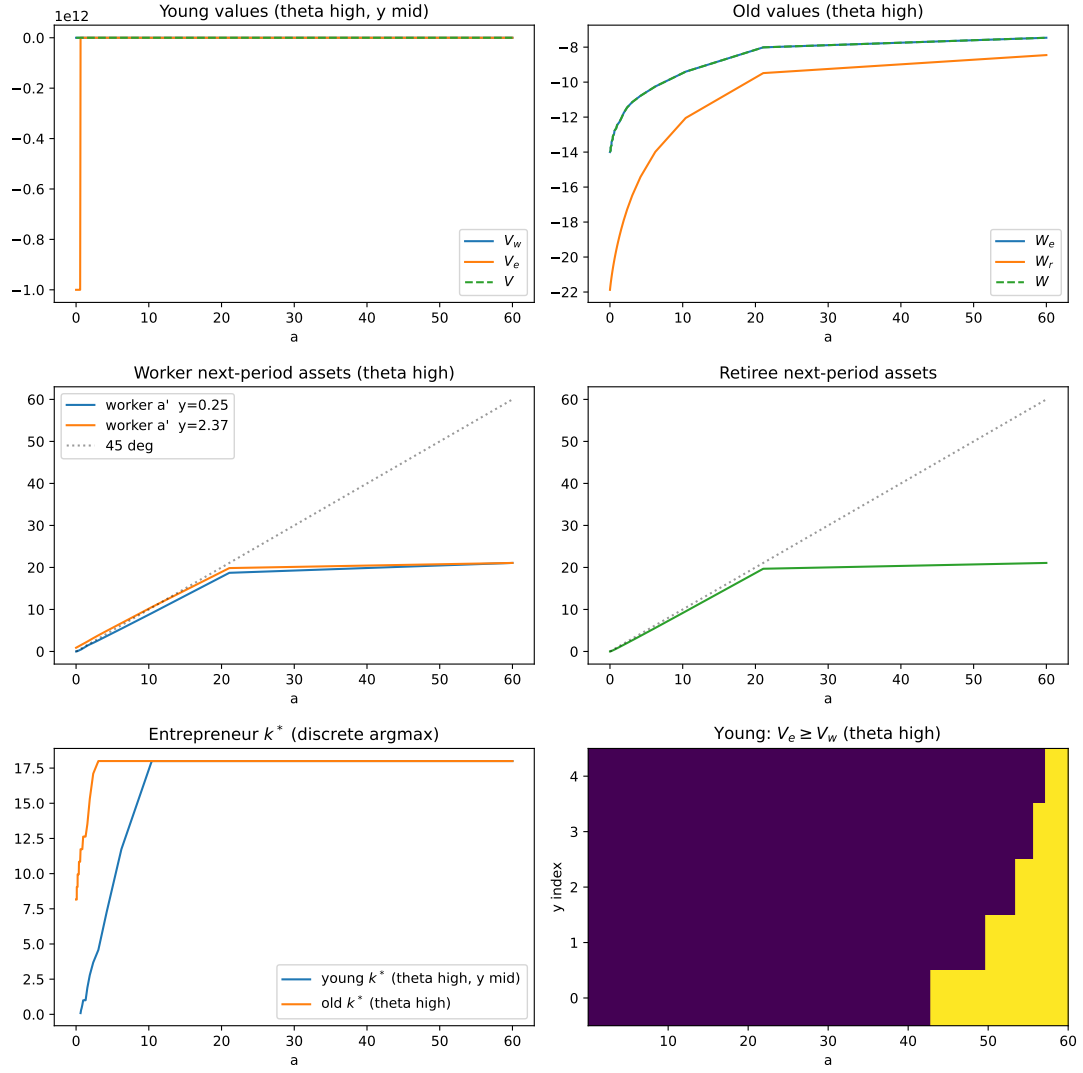


Figure 2 Value functions, saving policies, occupational choice, and entrepreneurial investment from the notebook.

The middle panels show next-period asset policies for workers and retirees. Both policies are below the 45-degree line at high assets in the plotted range, which means households do not keep increasing assets one-for-one forever. This is partly due to the finite grid and fixed-price setting. The bottom-left panel shows the entrepreneurial capital choice k^* . The old entrepreneur reaches a high business scale more quickly, while the young entrepreneur's k^* rises with assets and then flattens near the upper part of the discrete k grid. This is exactly the mechanism I wanted the reproduction to show: assets relax the ability to operate a larger firm, but the enforcement constraint and the discrete capital grid limit the feasible scale.

The bottom-right panel is the clearest occupational-choice picture. With zero entrepreneurial ability, nobody chooses entrepreneurship. With high entrepreneurial ability, the probability of choosing entrepreneurship is higher for lower labor productivity states and higher assets. In the current run, the share of the asset grid where $V_e \geq V_w$ is 28.7 percent for the lowest labor productivity state and 5.0 percent for the highest labor productivity state. This makes economic sense because high labor productivity raises the worker outside option.

6.3 Comparison with the Original Paper

Table 5 compares the notebook's main numerical outputs with the original paper's benchmark model results. The entrepreneur share is the closest comparison. The original paper reports 7.5 percent entrepreneurs in the baseline model, and the notebook's young-only distribution gives 8.25 percent. This is close enough to show that the occupational-choice mechanism is working in the right range.

Table 5 Benchmark Results and Current Reproduction Outputs

Moment	Original paper baseline	Notebook output	Interpretation
Entrepreneur share	7.50%	8.25%	Close in the fixed-price, young-only diagnostic
Wealth Gini	0.80	0.62	Too low because full demographic wealth accumulation is not reproduced
Top 1% wealth share	31%	6.10%	Far below paper; upper tail is missing
Top 5% wealth share	60%	27.78%	Below paper; young-only kernel understates concentration
Capital-output ratio	3.0	Not reproduced	Requires general equilibrium aggregation

The comparison shows both success and limitation. The success is that the notebook reproduces the branch structure well enough to generate an entrepreneur share near

the paper’s baseline. It also reproduces the qualitative occupational-choice logic: high entrepreneurial ability and sufficient assets make entrepreneurship more attractive, while high labor income makes wage work more attractive.

The limitation is the wealth distribution. The notebook’s wealth Gini is 0.62 and its top 1 percent wealth share is only 6.1 percent, far below the paper’s baseline values. I do not interpret this as evidence against the original model. Instead, it reflects the narrower reproduction. The original paper’s wealth concentration comes from the full distribution over ages, entrepreneurs, retirees, and dynasties, combined with general equilibrium and calibration. My notebook uses only a young-only diagnostic distribution, so it does not yet include the full bequest and old-age wealth accumulation channels that are important for the upper tail.

6.4 Stationary Distribution Diagnostic

Figure 3 reports the young-only distribution diagnostic. The right panel shows that the distribution iteration itself converges smoothly. The left panel shows mass both near zero assets and near the upper part of the grid. This tells me that the policy rules are moving households through the asset grid in a stable way. However, it also shows why the distribution comparison is limited. The diagnostic distribution is shaped by the truncated asset grid and does not include the full demographic status dimension, so it cannot fully reproduce the paper’s top wealth shares.

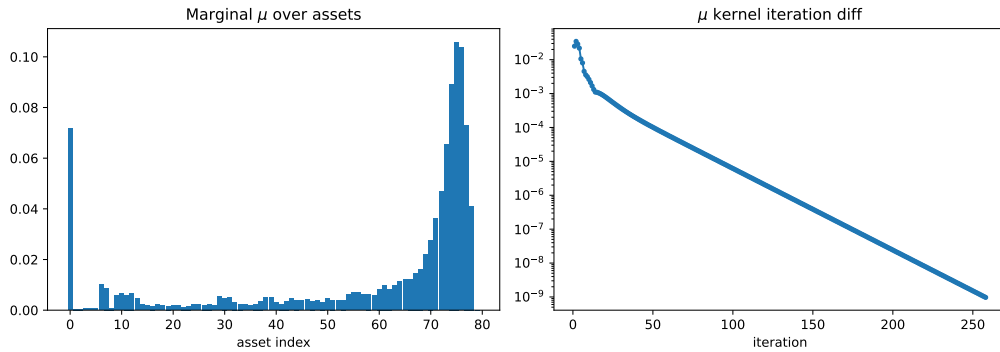


Figure 3 Young-only asset distribution diagnostic from the notebook.

The partial-equilibrium counterfactuals in the notebook should also be read cautiously. Some one-at-a-time changes, such as raising the interest rate to 7.5 percent or changing the pension share, converge and leave the middle-productivity young entrepreneurship region close to the baseline. Other changes, such as shutting down the bequest motive or increasing high entrepreneurial ability, do not converge within the shortened counterfactual iteration cap. I therefore use these rows only as diagnostics, not as reproduced versions of the paper’s counterfactual tables.

Overall, the reproduction is strongest on the household problem and weaker on the aggregate distribution. It succeeds at organizing the model, solving the baseline fixed-price problem, and showing the occupational-choice mechanism. It does not yet reproduce the

full wealth-concentration result, because that requires the full stationary equilibrium and demographic distribution.

7 Conclusions

This project reproduces the theoretical structure of the Cagetti and De Nardi household problem. The main contribution of my reproduction is organizational and computational: it restates the model, separates the worker and entrepreneur branches, solves the baseline fixed-price household problem, and compares the notebook output with the original paper’s benchmark moments.

The reproduction captures the central economic mechanism. Workers solve a standard consumption-saving problem, while entrepreneurs also choose business capital and must satisfy an enforcement constraint. Occupational choice compares the value of these two branches. In the baseline run, the young-only entrepreneur share is 8.25 percent, close to the original paper’s 7.5 percent benchmark. This suggests that the branch structure and occupational-choice logic are working in the right range.

The reproduction is much less successful for the wealth distribution. The notebook’s young-only wealth Gini is 0.62, compared with 0.80 in the paper’s baseline model, and the top 1 percent wealth share is much lower than in the original paper. This is not surprising because the notebook does not yet include the complete demographic stationary distribution, market clearing, full calibration, or the dynastic wealth accumulation channel.

The most natural next step would be to replace the young-only stationary distribution check with a full demographic distribution over young workers, young entrepreneurs, old entrepreneurs, and retirees. After that, the model could be closed with market clearing and social-security budget balance, and only then would it be appropriate to compare the reproduced wealth distribution and counterfactuals more directly with the original paper’s quantitative tables.

Appendices

A Income and Entrepreneurial Ability Processes

We assume that the income process is lognormal and AR(1). We approximate it with a five-point discrete Markov chain, using the method described in Tauchen and Hussey (1991).

The resulting grid points for the vector $\mathbf{y}$ for the income process (normalized to an average of one) are

$$[0.2468 \quad 0.4473 \quad 0.7654 \quad 1.3097 \quad 2.3742],$$

and the transition matrix $\mathbf{P}_y$ is

$$\mathbf{P}_y = \begin{bmatrix} .7376 & .2473 & .0150 & .0002 & .0000 \\ .1947 & .5555 & .2328 & .0169 & .0001 \\ .0113 & .2221 & .5333 & .2221 & .0113 \\ .0001 & .0169 & .2328 & .5555 & .1947 \\ .0000 & .0002 & .0150 & .2473 & .7376 \end{bmatrix}.$$

We assume that the entrepreneurial ability process is uncorrelated with the income process. The two values for ability θ are zero (no entrepreneurial ability) and a positive value (0.514), and the transition matrix $\mathbf{P}_\theta$ is

$$\mathbf{P}_\theta = \begin{bmatrix} .964 & .036 \\ .206 & .794 \end{bmatrix}.$$

A.1 Correlation between Abilities

We have assumed that working ability (y) and entrepreneurial ability (θ) are uncorrelated. One important piece of evidence in favor of this specification is that we replicate fairly well the income of entrepreneurs prior to starting their businesses. Using the PSID, the labor earnings over the previous five years of those who enter entrepreneurship in a given period are 1.32 times the labor income during the previous five years of those who choose to remain workers. Our simulations reproduce this feature: the ratio is 1.35. This correlation arises endogenously because of borrowing constraints—among the high-entrepreneurial ability workers, those who receive high labor earnings realizations can save more and are more likely to enter entrepreneurship.

As a robustness check, we study the effects of allowing for positive correlation of 0.4 between the two ability processes. The resulting wealth distribution is close to the benchmark: the richest 1 percent hold 33 percent of the total wealth. The main difference is that the number of entrepreneurs decreases to 5.2 percent because people with high entrepreneurial ability tend to have a higher option value of remaining in the nonentrepreneurial sector.

B The Algorithm

The algorithm proceeds as follows:

- Construct a grid for the state variables. The maximum asset level is chosen so that it is not binding.
- Fix a tax rate τ , an interest rate r , and a wage rate w . Taking these as given, solve for the value functions using value function iteration.
- Construct the transition matrix $\mathbf{M}$. Compute the associated invariant distribution, starting from a guess for $\boldsymbol{\pi}$ and iterating on $\boldsymbol{\pi}' = \mathbf{M}\boldsymbol{\pi}$ until convergence.
- Compute total savings and total capital invested in the entrepreneurial sector from the invariant distribution.
- Compute r and w implied by the nonentrepreneurial aggregate production function, and iterate until convergence on the factor prices.
- Compute the social security system imbalance and iterate on τ until outlays equal revenues.

The computation of the value functions is nonstandard because of the endogenous borrowing constraints. For each state $\mathbf{x}$, the endogenous constraint specifies a maximum amount $k(\mathbf{x})$ that can be borrowed, but this function depends on the value functions themselves. We initialize $\hat{k}(\mathbf{x}) = k_{\max}$, solve the value functions conditional on this constraint, then update $\hat{k}(\mathbf{x})$ by finding the maximum investment for which the entrepreneur would not default, and iterate until $\hat{k}$ converges.

C Additional Results (placeholder)

C.1 Introduction

C.2 Further details

References

- BUERA, FRANCISCO J (2006): “Persistency of Poverty, Financial Frictions, and Entrepreneurship,” .
- CAGETTI, MARCO, AND MARIACRISTINA DE NARDI (2006): “Entrepreneurship, frictions, and wealth,” *Journal of Political Economy*, 114(5), 835–870, SSRN 2005; JPE October 2006.
- EVANS, DAVID S, AND BOYAN JOVANOVIĆ (1989): “An Estimated Model of Entrepreneurial Choice,” *Journal of Political Economy*, 97(4), 808–827.
- GENTRY, WILLIAM M, AND R GLENN HUBBARD (2004): “Entrepreneurship and Household Saving,” *Advances in Economic Analysis and Policy*, 4(1).
- QUADRINI, VINCENZO (2000): “Entrepreneurship, Saving, and Social Mobility,” *Review of Economic Dynamics*, 3(1), 1–40.
- TAUCHEN, GEORGE, AND ROBERT HUSSEY (1991): “Quadrature-Based Methods for Obtaining Approximate Solutions,” *Econometrica*, 59(2), 371–396.